NUTRIFY

PoultAI & PoultSynapse

Calculations, discussion and explanation of the two broiler growth-prediction laboratories

What this document is. A complete, step-by-step account of every equation used in the two labs, with worked numerical examples produced by running the real website code, a discussion of what each calculation means biologically, and an honest list of what is reference data and what is a modelling assumption. It is written so that a new team member can understand, check and improve the models.

Lab	Folder	Purpose
PoultAI (v1)	poultai/	Simple 4-nutrient model: energy, crude protein, lysine, methionine + cystine, plus temperature.
PoultSynapse (v2)	poultsynapse/	Advanced model: 32 Cobb nutrient factors, sex-specific day-by-day tables, feeding phases, amino acid balance, heat stress. Formerly the folder poultai-2.

Reading guide: Part 1 explains the shared ideas. Part 2 documents PoultAI. Part 3 documents PoultSynapse. Part 4 compares them. Part 5 explains how to extend the code. All numbers in the worked examples are taken from the code, so you can reproduce them by moving the sliders.

Part 1 · Shared ideas

1.1 The one-sentence idea

Both labs answer the same question: **if a Cobb 500 broiler is fed a ration that differs from the Cobb recommendation, how much lighter (or heavier) will it be at a given age?** The answer is built from a published reference weight multiplied by a *performance coefficient* that describes how good the ration is.

```
predicted weight  =  Cobb target weight (age)  ×  performance coefficient
variance (g)      =  predicted weight  −  Cobb target weight
variance (%)      =  variance / Cobb target weight × 100
performance coefficient  = 1.000  → the ration meets Cobb targets (bird on the Cobb curve)
                         < 1.000  → ration, heat or imbalance holds growth back
                         > 1.000  → small gain from feeding above target (v2 caps this)
```

1.2 What comes from Cobb and what is assumed

Item	Status	Explanation
Target live weight, FCR, daily feed intake	**Published**	Cobb500 Broiler Performance & Nutrition Supplement (2022). v2 stores every day 0–56 for as-hatched, male and female; v1 stores seven days and interpolates.
Nutrient targets per phase (energy, protein, amino acids, minerals, linoleic acid, vitamins, trace elements)	**Published**	Same Cobb supplement. Ranges (e.g. sodium 0.16–0.23 %) are kept as ranges in v2.
Ideal amino-acid ratios (% of lysine)	**Published**	Cobb "balanced protein" profile (v2 only).
How strongly each nutrient shortfall reduces weight (weights, sensitivities)	**Assumption**	Cobb publishes requirements, not response curves. These numbers are our hypothesis and the main thing to calibrate with trial data.
Heat-stress penalty (1.8 % per °C above 25 °C)	**Assumption**	Kept from the original PoultAI. Direction is realistic; the size is not validated.
Chicken drawing size and colour	**Illustration**	Purely visual; it follows the coefficient.

1.3 Reading a prediction responsibly

A prediction is a **hypothesis generator**, not a forecast. It is useful for asking "which nutrient matters most here?" and for teaching. It is not a substitute for veterinary assessment, professional feed formulation or a real feeding trial. The labs say so on screen and the Home page repeats it.

Part 2 · PoultAI (v1) — the four-nutrient model

Source file: `poultai/poultai.js`. Inputs: bird age (1–42 days), metabolizable energy (kcal/kg), crude protein (%), digestible lysine (%), methionine + cystine (%), and ambient temperature (°C).

2.1 Step 1 — Cobb target weight by interpolation

The code stores the Cobb as-hatched weight for seven ages: day 1 = 42 g, 7 = 202 g, 14 = 512 g, 21 = 1116 g, 28 = 1782 g, 35 = 2521 g, 42 = 3278 g. For any age between two stored days it draws a straight line:

```
target(d) = w1 + (d − d1) / (d2 − d1) × (w2 − w1)        d1 ≤ d ≤ d2, result rounded to 1 g

Example, day 10 (between day 7 and day 14):
target(10) = 202 + (10 − 7)/(14 − 7) × (512 − 202) = 202 + 0.4286 × 310 = 335 g
```

2.2 Step 2 — Phase nutrient targets

Phase (days)	Energy kcal/kg	Crude protein %	Dig. lysine %	Methionine (+cys) %
Starter (0–10)	3000	21.5	1.22	0.90
Grower (11–22)	3050	19.5	1.10	0.82
Finisher (23+)	3150	18.0	1.00	0.76

2.3 Step 3 — Ratios and the performance coefficient

Each nutrient is divided by its target to give a ratio (1.00 = on target). The coefficient is a fixed-weight average of the four ratios, multiplied by a temperature index.

```
ratio_x          = input_x / target_x                (x = energy, protein, lysine, methionine)
temperature index = 1                                 if T ≤ 25 °C
                  = 1 − (T − 25) × 0.018              if T > 25 °C
coefficient = (0.35·r_energy + 0.28·r_protein + 0.22·r_lysine + 0.15·r_met) × temperature index
  (r = ratio)
weights: 0.35 + 0.28 + 0.22 + 0.15 = 1.00
predicted weight = round( target × coefficient )
```

2.4 Step 4 — Feed conversion ratio and the chicken

```
FCR (estimated) = 1.18 × target weight / predicted weight         (shown with 2 decimals)
chicken scale   = age scale × nutrient scale
  age scale     = 0.45 + (age / 42) × 0.75
  nutrient scale = coefficient limited to the range 0.70 … 1.25
phenotype label: coefficient < 0.85 → "Deficient"; > 1.08 → "High-Density"; otherwise "Optimal"
```

2.5 Worked examples (values produced by the site code)

Case (day 21 unless stated)	Inputs	Coeff.	Predicted	Target	Var.	FCR
A. Standard ration	3050 / 19.5 / 1.10 / 0.82, 22 °C	1.000	1116 g	1116 g	0 g	1.18
B. "Deficient scenario" button	2700 / 16.5 / 0.85 / 0.60, 28 °C	0.782	873 g	1116 g	-243 g	1.51
C. Standard ration, heat only	as A, 32 °C	0.874	975 g	1116 g	-141 g	1.35
D. Energy 20 % above target	3660 / 19.5 / 1.10 / 0.82, 22 °C	1.070	1194 g	1116 g	+78 g	1.10

Case B in detail. Ratios: energy 2700/3050 = 0.885; protein 16.5/19.5 = 0.846; lysine 0.85/1.10 = 0.773; methionine 0.60/0.82 = 0.732. Weighted sum = 0.35×0.885 + 0.28×0.846 + 0.22×0.773 + 0.15×0.732 = 0.8383. Temperature index at 28 °C = 1 − 3×0.018 = 0.946. Coefficient = 0.8383 × 0.946 = 0.7819 → 1116 × 0.7819 = 873 g (−243 g, −21.8 %). FCR = 1.18 × 1116/873 = 1.51.

2.6 Discussion — strengths and limits of v1

- **Strength:** transparent. Four sliders, one line of arithmetic; students can verify every number by hand.
- **Linear and unbounded:** because the coefficient is a plain weighted average of ratios, feeding more always gives more weight, with no ceiling and no penalty for excess. Case D shows +7 % weight from 20 % extra energy, which real birds would not deliver (surplus energy is stored as fat or wasted). PoultSynapse fixes this with a capped bonus and an excess penalty.
- **Nutrients are averaged, not limiting:** in biology the most deficient essential nutrient limits growth. A 22 % lysine shortfall is partly hidden by an on-target energy ratio in v1. v2 adds a "limiting nutrient" term.
- **No amino-acid balance:** v1 does not know that methionine must follow lysine in a fixed proportion.
- **Constant FCR base (1.18):** real Cobb FCR rises with age (0.89 at day 7 to 1.84 at day 56 in the as-hatched table); v2 uses the real value for the day.
- **Reference-table check (action for the team):** the day-14 weight stored in v1 is 512 g and the day-1 value is 42 g, whereas the day-by-day Cobb table used in v2 lists 570 g at day 14 and 55 g at day 1 (42 g is the day-0 weight). Please compare with the Cobb PDF and update v1 if you agree. This document did not change v1's numbers.

Part 3 · PoultSynapse (v2) — the full-spectrum model

Source file: `poultsynapse/poultsynapse.js` (renamed from poultai-2/poultai.js). The file is organised in numbered sections: 0 theme colours, 1 reference data, 2 lookup helpers, 3 prediction model (pure maths, no page access), 4 interface builders, 5 display updaters, 6 scenarios and events, 7 Dr. Synapse comments. Inputs: age (1–56 d), sex (as hatched / male / female), market (medium-large 5 phases / small 4 phases), cereal base (maize / wheat), temperature, and **32 nutrient levels**.

3.1 Reference lookups

- **Target weight:** exact table lookup, `COBB_PERFORMANCE[sex].weight[day]` (no interpolation needed because Cobb lists every day). Example: male, day 35 → 2694 g.
- **Target FCR and daily feed intake** are read the same way (FCR from day 7, intake from day 8).
- **Feeding phase:** the first phase whose last day is ≥ the bird's age. Medium/large programme: Starter 0–12, Grower 1 13–28, Grower 2 29–39, Finisher 1 40–49, Finisher 2 50+. Small-bird programme: Starter 0–8, Grower 1 9–18, Grower 2 19–28, Finisher 29+.
- **Nutrient target:** macro nutrients use targets[market][phase]; vitamins and trace elements use the premix column (Starter / Grower / Finisher), and biotin switches between maize and wheat values. A target can be a single number or a range [low, high]; the slider is reset to the range midpoint.

3.2 Step 1 — Scoring one nutrient

Every nutrient receives a score. A score of 1.00 means "meets the Cobb target". Let *lo* and *hi* be the target range (equal for a single value), *v* the chosen level, and the group parameters sens, bonus, excessAt, excessSens (Table 3.3).

```
if v < lo :   score = max( 0 ,  1 − sens × (lo − v) / lo )        ← shortfall penalty
if lo ≤ v ≤ hi :  score = 1                                       ← inside the target range
if v > hi :   rel   = (v − hi) / hi                                  ← relative excess
              gain  = bonus × min(rel, 0.08) / 0.08                  ← up to +bonus at +8 %
              pen   = excessSens × max(0, rel − excessAt)            ← harmful-excess penalty
              score = max( 0 ,  1 + gain − pen )
```

Example: lysine target at day 21 = 1.16 %. At 0.99 % (−14.7 %) the amino-acid group has sens = 1.0, so score = 1 − 1.0 × (1.16 − 0.99)/1.16 = 0.853. At 1.25 % (+7.8 %) the gain is 0.04 × 0.078/0.08 = +0.039 and no penalty, so score = 1.039.

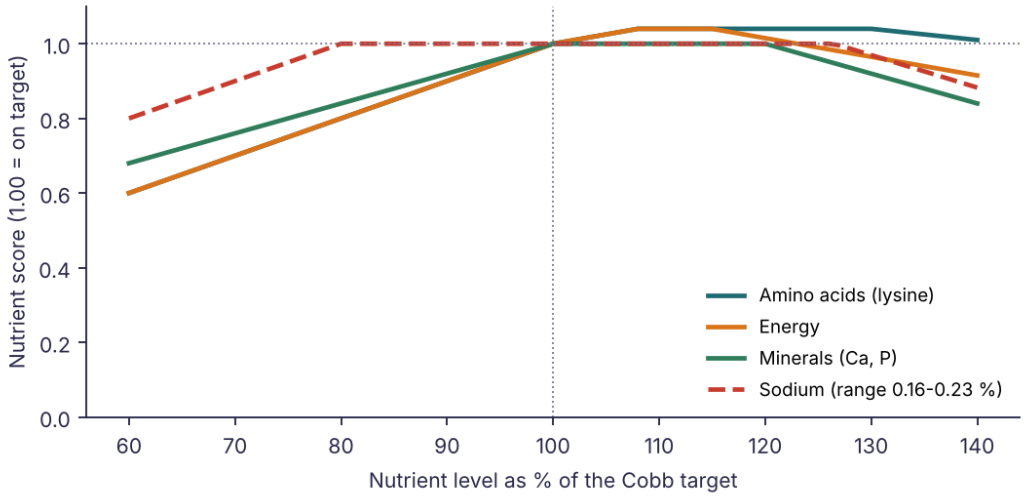


Figure 1. Nutrient score against level (% of target) for four nutrient types. Energy and amino acids can gain a little above target; minerals cannot gain but are penalised sooner; sodium has a range and an early excess penalty.

3.3 Nutrient groups (model parameters)

Group	Weight	sens	bonus	excessAt	excessSens	Members
Energy	0.25	1	0.04	0.15	0.5	1
Crude Protein	0.15	1	0.04	0.25	0.4	1
Digestible Amino Acids	0.30	1	0.04	0.3	0.3	9: lysine, Met+Cys, Met, Trp, Thr, Arg, Val, Ile, Leu
Minerals & Electrolytes	0.12	0.8	0	0.2	0.8	5: Ca, available P, Na, Cl, K
Essential Fatty Acid	0.03	0.5	0	0.5	0.2	linoleic acid
Supplementary Vitamins (per tonne)	0.08	0.4	0	0.5	0.2	13: A, D3, E, K, B1, B2, B6, B12, biotin, choline, folic, nicotinic, pantothenic
Supplementary Trace Elements (per tonne)	0.07	0.4	0	0.5	0.4	6: Mn, Zn, Fe, Cu, I, Se

The seven weights add up to 1.00. Inside a group each nutrient has its own importance weight (default 1; lysine 3, Met+Cys 2, methionine 1.5, threonine 2, calcium 2, available phosphorus 2). Sodium, chloride and potassium override excessAt = 0.10 and excessSens = 1.0 because they are ranges where excess is harmful.

3.4 Step 2 — Combine the scores

```
group score_g   = Σ (importance_i × score_i) / Σ importance_i              (nutrients i in group g)
weighted score  = Σ_g  group weight_g × group score_g                        (seven groups)
limiting score  = min( 1 , lowest single nutrient score )
nutrient coefficient = (1 − 0.20) × weighted score + 0.20 × limiting score
```

The 20 % limiting share encodes Liebig's idea that the scarcest essential nutrient restricts growth: even if most scores are good, one very low score drags the coefficient down.

3.5 Step 3 — Amino-acid balance penalty

Cobb gives the ideal amount of each amino acid as a percentage of lysine (e.g. Met+Cys 75–77 %, threonine 65–68 %, arginine 108–110 %). The model compares the ratio the user actually creates with the ideal ratio for the phase and averages the relative error over the eight amino acids:

```
actual_a = level_a / level_lysine × 100            ideal_a = Cobb ratio for the phase
average error = mean over 8 amino acids of  |actual_a − ideal_a| / ideal_a
balance penalty = min( 0.06 ,  0.25 × max(0, average error − 0.05) )          (never more than 6 %)
```

Why it matters: raising lysine alone makes every other ratio too small. In Figure 3 and Table 3.7 a +8 % lysine change alone lowers the result slightly instead of raising it.

3.6 Step 4 — Heat stress, final weight, FCR

```
temperature index = 1                                if T ≤ 25 °C
                  = max( 0.5 , 1 − 0.018 × (T − 25) )   if T > 25 °C
performance coefficient = nutrient coefficient × (1 − balance penalty) × temperature index
predicted weight = round( Cobb target(age, sex) × performance coefficient )
predicted FCR    = Cobb FCR(age, sex) × Cobb target weight / predicted weight     (3 decimals)
```

The FCR formula says: a bird that is only 80 % of its target weight has eaten roughly the feed of a normal bird, so feed per kg of gain rises by 1/0.80. It is an estimate, not a Cobb figure.

3.7 Worked examples (values produced by the site code)

Case (day 21, as hatched, medium, maize)	Coeff.	Predicted	Variance	FCR	Notes
A. Standard ration (all targets at range mid-point, 22 °C)	1.000	1116 g	0 g (0.0 %)	1.182	Reproduces Cobb exactly (1116 g, FCR 1.182).
B. "Deficient Scenario" button	0.805	898 g	-218 g (-19.5 %)	1.469	Amino acids 78 %, minerals 90 %, vitamins/trace 70 %, energy 2700, CP 16.5 %, 28 °C. Limiting score 0.78: every amino acid sits at 78 %.
C. Lysine +8 % only	0.996	1111 g	-5 g (-0.4 %)	1.187	Imbalance: average ratio error 7.5 % → penalty 0.6 %.
D. Lysine −15 % only	0.932	1040 g	-76 g (-6.8 %)	1.268	Limiting nutrient lysine (score 0.850); balance penalty 3.1 %.
E. Calcium −20 % only	0.964	1075 g	-41 g (-3.7 %)	1.227	Calcium score 0.84 (minerals sens 0.8 × 20 % = 16 % loss in that nutrient).
F. Heat 32 °C only	0.874	975 g	-141 g (-12.6 %)	1.353	Temperature index 1 − 7 × 0.018 = 0.874.

Case B, full calculation. Group scores from the code: energy 0.915, crude protein 0.868, amino acids 0.780, minerals 0.954, fatty acid 0.950, vitamins 0.880, trace elements 0.880.

```
weighted score        = 0.25×0.915 + 0.15×0.868 + 0.30×0.780 + 0.12×0.954
                        + 0.03×0.950 + 0.08×0.880 + 0.07×0.880 = 0.8681
limiting score        = 0.780  (all amino acids at 78 %)
nutrient coefficient  = 0.80 × 0.8681 + 0.20 × 0.780 = 0.8505
balance penalty       = 0.000   (average ratio error 0.8 % is inside the 5 % tolerance)
temperature index     = 1 − 3 × 0.018 = 0.946
coefficient           = 0.8505 × (1 − 0.000) × 0.946 = 0.8045
predicted weight      = 1116 × 0.8045 = 898 g        FCR = 1.182 × 1116 / 898 = 1.469
```

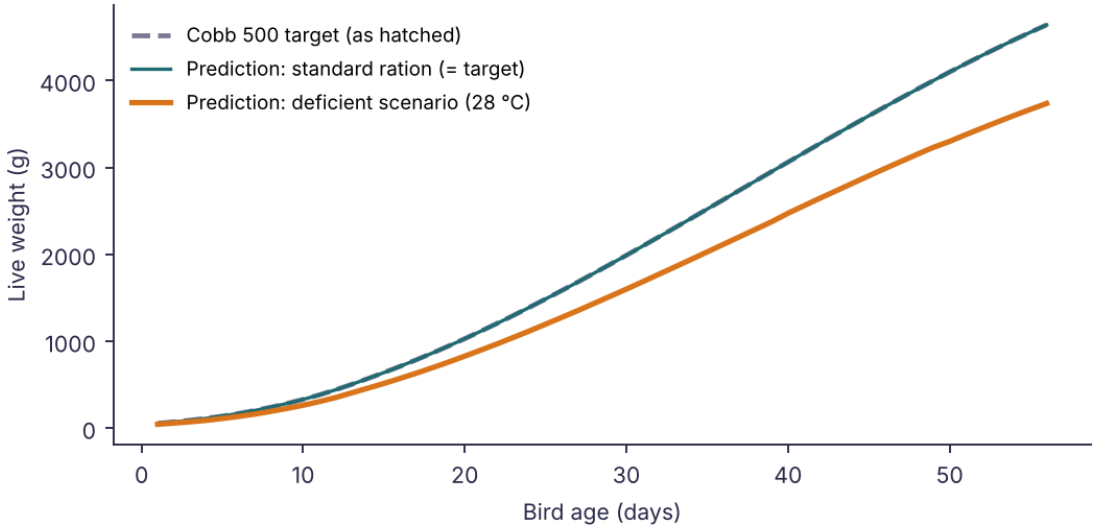


Figure 2. Growth curves from day 1 to 56. The deficient ration is applied on every day of the curve (each day uses that day's own Cobb targets scaled by the same factors).

3.8 Status labels, chicken and Dr. Synapse

- **Table status:** within ±2 % of the Cobb range = OPTIMAL; below = DEFICIENT; above = SURPLUS. Deviation shown is 0 inside the range, otherwise the distance to the nearest bound in %.
- **Chicken:** scale = (0.55 + age/56 × 0.65) × clamp(coefficient, 0.70, 1.25). Label "Deficient" below 0.85, "High-Density" above 1.015, otherwise "Optimal". Colours come from CSS variables.
- **Dr. Synapse:** the speech bubble picks the first matching rule in `DOCTOR_LINES` (heat, severe loss, deficiency, imbalance, above-Cobb, on-target). Add a line to the list to add a remark.
- **Chart:** for every day 1–56 the model is run with the current ration; the grey dashed curve is the Cobb table, the solid curve the prediction, and the dot marks the selected age.

3.9 Discussion — what the model does well and what it assumes

- **Bounded behaviour:** bonuses are capped at +4 % for energy, protein and amino acids and zero elsewhere; excess is penalised. The coefficient cannot run away as it can in v1.
- **Interactions:** amino-acid imbalance, limiting nutrient and heat stress all act on top of the weighted average, so combining problems is worse than the sum of mild cases.
- **Cereal and phase awareness:** targets change with the programme, the phase and the cereal (biotin), which is how a nutritionist would read the Cobb tables.
- **Assumption 1 — sensitivities.** All sens / bonus / excess values are hypotheses. The easiest way to improve realism is to fit them to trial data from the faculty or the literature.
- **Assumption 2 — same relative effect at every age.** A fixed fractional loss is applied to the target weight at each age; real deficits compound over time and can partly be recovered (compensatory growth).
- **Assumption 3 — nutrients act independently** (except through the balance penalty and the limiting term). Real interactions (calcium–phosphorus ratio, electrolyte balance, vitamin E–selenium) are not modelled.
- **Not modelled:** stocking density, ventilation, litter, disease, water quality, feed form quality and feed intake itself. These are listed on the PoultSynapse page as real-life factors. One Cobb 500 trial (Shewita et al., 2019) found lower final weight, lower feed intake and higher mortality at 15.6 vs 10.6 birds/m², illustrating that housing can outweigh formulation.

Part 4 · PoultAI versus PoultSynapse

Aspect	PoultAI (v1)	PoultSynapse (v2)
Nutrient factors	4 (energy, CP, lysine, Met+Cys)	32 (energy, CP, 9 amino acids, 5 minerals, linoleic acid, 13 vitamins, 6 trace elements)
Age range / weight table	1–42 d; 7 stored points, linear interpolation	0–56 d; every day from Cobb, three sex tables
Feeding phases	3 fixed phases	Cobb medium-large (5) or small-bird (4) programme; cereal-specific biotin
Combining rule	Fixed weighted average of ratios	Per-nutrient scores → group means → weighted sum + 20 % limiting nutrient
Excess feeding	Always increases weight (unbounded)	Capped bonus (≤ +4 %) then penalty
Amino-acid balance	Not modelled	Ratios to lysine vs Cobb profile, penalty ≤ 6 %
FCR	1.18 × target / predicted	Cobb FCR of the day × target / predicted
Heat stress	−1.8 % per °C above 25 °C	Same, floored at 0.5
Look	Dark neon lab	Light, Van Gogh-inspired mad-scientist lab with Dr. Synapse

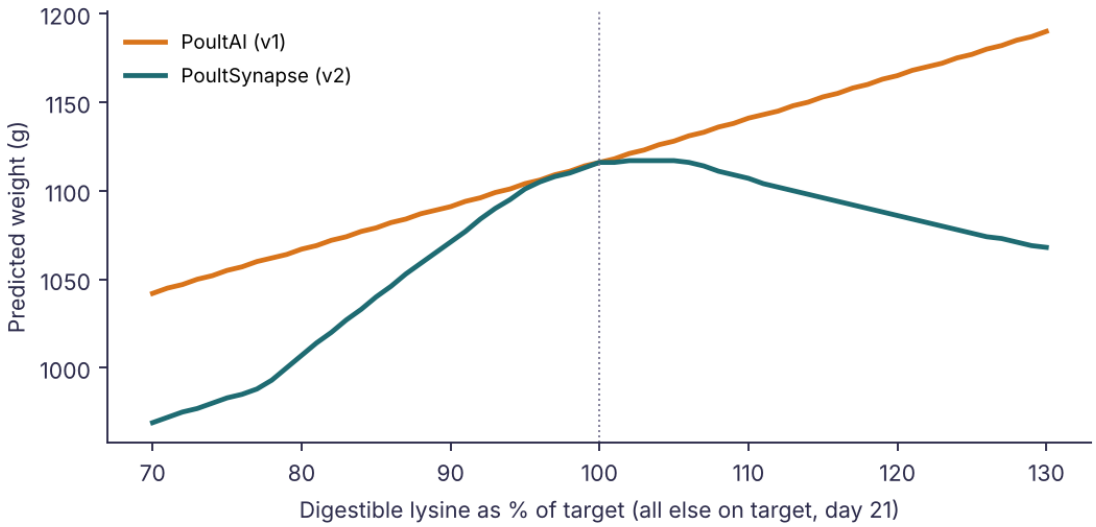


Figure 3. Same ration, only lysine varied. v1 rises steadily without limit; v2 peaks near the target because the other amino acids then become relatively short (imbalance) and the bonus is capped.

Reading Figure 3: at 120 % lysine v1 predicts 1165 g while v2 predicts 1086 g. Biologically, extra lysine with no matching increase in the other amino acids cannot be used for growth, so v2's behaviour is the more defensible one. At 70 % lysine v1 gives 1042 g and v2 gives 969 g; both penalise a shortage.

Part 5 · Extending the code

I want to…	Where
Add a nutrient to PoultSynapse	Add one block to `NUTRIENTS` (id, label, unit, group, slider limits, targets). The slider, matrix row and score appear automatically.
Change a sensitivity, weight or heat penalty	`NUTRIENT_GROUPS` and `MODEL_CONSTANTS` at the top of poultsynapse.js.
Restyle PoultSynapse	The `:root` block of poultsynapse.css (colours, fonts, shapes). Chart colours follow the --chart-* variables.
Change what Dr. Synapse says	`DOCTOR_LINES` at the end of poultsynapse.js.

I want to…	Where
Add a paper	Copy a block in PAPERS in js/content.js. The card, filter chip and Copy-citation button are created automatically with the shared CSS classes.
Add a tip	Copy a block in TIPS in js/content.js.
Change main site colours	The `:root` block of css/style.css; new components inherit them.

Part 6 · References

- Cobb-Vantress. *Cobb500 Broiler Performance & Nutrition Supplement* (2022). https://cobbgenetics.com/assets/Cobb-Files/2022-Cobb500-Broiler-Performance-Nutrition-Supplement.pdf
- Shewita RS, El-Naggar K, Abd El Naby WS. Influence of dietary vitamin C supplementation on growth performance, blood biochemical parameters and transcript levels of heat shock proteins in high stocking density reared broiler chickens. *Slovenian Veterinary Research* 56 (Suppl. 22), 2019. DOI 10.26873/SVR-750-2019.
- Yu DG, Namgung N, Kim JH, Won SY, Choi WJ, Kil DY. Effects of stocking density and dietary vitamin C on performance, meat quality, intestinal permeability, and stress indicators in broiler chickens. *J Anim Sci Technol* 2021. DOI 10.5187/jast.2021.e77.
- National Research Council. *Nutrient Requirements of Poultry*. National Academies Press (listed on the site's References tab).

Part 7 · Change log of this version

- **Mobile navigation fixed** in both labs and the main site: headers wrap, grid panels can shrink, nothing scrolls sideways; main menu now collapses at 980 px, closes after a tap, outside click or Escape.
- **New tabs:** PoultSynapse (renamed from poultai-2, all names and files changed), Tips, Papers. Tips and papers are generated from js/content.js.
- **PoultSynapse redesign:** light Van Gogh-inspired mad-scientist theme, mascot, handwritten notes, live result bar on small screens.
- **Code clean-up:** inline styles moved to CSS classes, drawing colours read from CSS variables, an unclosed table card in the PoultSynapse page fixed, comments added.
- **Not changed:** the NUTRIFY brand colours and logo, and the mathematical model of PoultAI v1 and PoultSynapse v2 (only renamed).